Question

Which of the following statement is not correct regarding cathode rays?

A. Cathode rays originate from the cathode.

B. Charge and mass of the particles constituting cathode rays depends up on the nature of the gas.

C. Charge and mass of the particles constituting cathode rays does not depends up on the material of the cathode.

D. They possess kinetic energy.

Question

The e/m ratio of cathode rays is x unit, when hydrogen is filled in the discharge tube. What will be its value when deuterium (D_2) is filled in it?

A. x unit

B. 2x unit

C. x/2 unit

D. x/4 unit

Question

The charge to mass ratio of electron was found to be:

A. 1.6022×10^{-19} C kg^{-1}

B. 1.925×10^{12} C kg^{-1}

C. 1.758×10^{11} C kg^{-1}

D. 1.869×10^{13} C kg^{-1}

Question

The e/m ratio of anode rays produced in the discharge tube, depends on the?

A. Nature of the gas filled in the tube

B. Nature of anode material

C. Nature of cathode material

D. All of these

Question

Gases are bad conductors of electricity. Their conductivity may be increased by

A. Increasing the pressure as well as potential difference between the electrodes.

B. Decreasing the pressure as well as potential difference between the electrodes.

C. Decreasing the pressure and/or increasing the potential difference between th electrodes.

D. Increasing the pressure and/or decreasing the potential difference between th electrodes.

Question

The specific charge of cathode rays

A. Depends on the nature of the gas.

B. Depends on the material of the discharge tube.

C. Depends on the potential difference between cathode and anode.

D. Is a universal constant

Question

The ratio of mass of an electron to that of the mass of hydrogen atom is:

A. 1 : 3871

B. 1 : 1837

C. 1 : 1296

D. 1 : 3781

Question

Millikan's oil drop experiments is used to find

(A) e/m ratio of an electron

(B) Charge on an electron

(C) Mass of an electron

(D) Velocity of an electron

Question

Select the sub-atomic particles that are present in the nucleus of an atom.

(1) Electrons **(2) Protons** **(3) Neutrons**

A. Only 2

B. Only 3

C. Both 1 and 2

D. Both 2 and 3

Question

According to Thomson, the amount of deviation of the particles from their path in the presence of electrical or magnetic field depends upon

A. the magnitude of the negative charge on the particle

B. the mass of the particle

C. the strength of the electrical or magnetic field

D. All of the above

Question

The e/m ratio of anode rays produced in the discharge tube, depends on the?

A. Nature of the gas filled in the tube

B. Nature of anode material

C. Nature of cathode material

D. All of these

Question

Positive rays or canal rays are:

A. Electromagnetic waves

B. A stream of positively charged gaseous ions

C. A stream of electrons

D. Neutrons

Question

Statement-I: The charge to mass ratio of the particles in anode rays depends on nature of the gas taken in the discharge tube.

Statement-II: The particles in anode rays carry positive charge.

A. Both Statement-I and Statement-II are correct.

B. Both Statement-I and Statement-II are incorrect

C. Statement-I is correct and Statement-II is incorrect

D. Statement-I is incorrect and Statement-II is correct

Question

What is the other name for anode rays?

A. Proton rays

B. Cation rays

C. Thomson rays

D. Canal rays

Question

What are the essential conditions for the production of anode rays?

A. High voltage and low pressure

B. High voltage and high pressure

C. Low voltage and high pressure

D. Low voltage and low pressure

Question

Which of the following is not true for cathode rays?

A. Cathode rays consist of negatively charged particles.

B. Cathode rays were discovered by J.J Thomson.

C. Nature of cathode rays depend upon the nature of the gas in discharge tube.

D. Cathode rays are produced inside the discharge tube.

Question

Which of the following statements about the electron is incorrect?

A. It is a negatively charged particle.

B. The mass of electron is equal to the mass of neutron.

C. It is a basic constituent of all atoms.

D. It is a constituent of cathode rays

Question

The e/m ratio is maximum for

A. Na^{+}

B. Al^{3+}

C. H^{+}

D. Mg^{2+}

Question

α-particles are represented by

A. Lithium atoms

B. Helium nuclei

C. Hydrogen nuclei

D. None of these

Question

Which is true?

A. e/m values of cathode rays generated by discharge of different gases are differen

B. (e/m) value of anode rays generated by discharge of different gases is same

C. $\left(\frac{e}{m}\right)_{\text{proton}} > \left(\frac{e}{m}\right)_{\text{electron}}$

D. $\left(\frac{e}{m}\right)_{\text{proton}} > \left(\frac{e}{m}\right)_{\alpha-\text{partide}}$

Question

Which of the following properties of atom could be explained correctly by Thomson Model of atom?

A. Overall neutrality of atom.

B. Spectra of hydrogen atom.

C. Position of electrons, protons and neutrons in atom.

D. Stability of atom.

Question

Which of the following models are not the same as Thomson model of Atom?

A. Plum Pudding model

B. Watermelon model

C. Raisin Pudding model

D. Nuclear model

Question

α-particles are represented by

A. Lithium atoms

B. Helium nuclei

C. Hydrogen nuclei

D. None of these

Question

The increasing order for the values of e/m (charge/mass) is:

A. e, p, n, α

B. n, p, e, α

C. n, p, α, e

D. n, α, p, e

Question

According to the electromagnetic theory of Maxwell, which one is correct?

A. Charged particles when accelerated should emit electromagnetic radiation

B. Charged particles when accelerated should absorb electromagnetic radiation

C. Charged particles when retarted should emit EMR

D. None of the above

Question

Which of the following substances, shows the properties of phosphorescent?

A. Zinc sulphide

B. Zinc sulphate

C. Zinc nitrate

D. Zinc chloride

Question

For $^{19}F^{-1}$, $^{16}O^{-2}$ ^{20}Ne, choose the correct statement.

A) Both O^{-2} and F^{-} are isoelectronic

B) All the given species have equal number of e^{-}

C) F^{-} and Ne have equal number of e^{-}

D) All of the above

Question

Mass number of an atom (A) is equal to

A. total number of nucleons

B. total number of neutrons

C. total number of protons and electrons

D. total number of protons, electrons and neutrons

Question

Cathode rays have same charge to mass ratio as

A. β-rays

B. α-rays

C. anode rays

D. None of the above

Question

The radius of nucleus is approximately ____times smaller than the radius of atom.

A. 1,00,000

B. 5,000

C. 10,000

D. 200

Question

When a gold sheet is bombarded by a beam of α-particles, only a few of them get deflected whereas most go straight, undeflected. This is because

A. The force of attraction exerted on the α- particle by the oppositely charged electrons is not sufficient.

B. A nucleus has a much smaller volume than that of atom

C. The force of repulsion acting on the fast moving α-particles is very small.

D. The neutrons in the nucleus do not have any effect on the α-particles

Question

Rutherford's experiment which established the nuclear model of the atom, used a beam of

A. β-particles, which impinged on a metal foil and got absorbed.

B. γ-rays, which impinged on a metal foil and ejected electrons.

C. helium atoms, which impinged on a metal foil and got scattered.

D. helium nuclei, which impinged on a metal foil and got scattered.

Question

The fraction of volume occupied by the nucleus with respect to the total volume of an atom is:

A. 10^{-15}

B. 10^{-5}

C. 10^{-30}

D. 10^{-10}

Question

Radius of nucleus varies as $R = R_0 (A)^{1/3}$ where $R_0 = 1.3$ Fermi. What is the volume of Be^8 nucleus (approx.) [A = atomic mass]?

A. 7×10^{-38} cc

B. 7×10^{-29} cc

C. 7×10^{-45} cc

D. None of these

Question

A particular station of All India Radio, New Delhi, broadcasts on a frequency of 1,368 kHz (kilohertz). The wavelength of the electronmagnetic radiation emitted by the transmitter is: [speed of light, $c = 3.0 \times 10^8\ ms^{-1}$]

A) 219.3 m

B) 219.2 m

C) 2192 m

D) 21.92 cm

Question

Which of the following have maximum frequency?

A. Cosmic rays

B. γ – rays

C. Micro waves

D. Radio waves

Question

The correct order of increasing frequency of electromagnetic radiations is:

A. Microwave < UV < IR < γ-rays

B. Microwave < Radiowave < Visible < X-rays

C. UV < Radiowave < X-rays < γ-rays

D. Radiowave < IR < Visible < X-rays

Question

Which of the following statement is/are correct about Hertz (Hz)?

A. It is the SI unit of frequency (v)

B. It is named after Heinrich Hertz

C. It is the number of waves that pass a given point in one second

D. All of the above

Question

Calculate the frequency and energy of a photon of wave length 5000 Å.

Question

Calculate the Wavelength and frequency of a photon having an energy of 2 electron volt.

Question

Calculate the energy in joule corresponding to light of wavelength 45 nm (Planck's constant, $h = 6.63 \times 10^{-34}$ Js; speed of light, $c = 3 \times 10^{8}$ ms^{-1}).

(A) 6.67×10^{15}

(B) 6.67×10^{11}

(C) 4.42×10^{-15}

(D) 4.42×10^{-18}

Question

The photons of light having a wavelength 4000 Å are necessary to provide 1.00 J of energy are:

A. 6.023×10^{23}

B. 6.023×10^{18}

C. 2.01×10^{18}

D. 2.01×10^{23}

Question

The value of Planck's constant is 6.63×10^{-34} Js. The speed of light is 3×10^{17} nm s^{-1}. which value is closest to the wavelength in nanometer of a quantum of light with frequency of 6×10^{15} s^{-1}?

A. 10

B. 25

C. 50

D. 75

Question

The energy of a radiation of wavelength 8000 Å is E_1 and energy of a radiation of wavelength 16000 Å is E_2. What is the relation between these two

A. $E_1 = 6E_2$

B. $E_1 = 2E_2$

C. $E_1 = 4E_2$

D. $E_1 = ½\ E_2$

Question

A near UV photon of 300 nm is absorbed by a gas and then re-emitted as two photons. One photon is red with the wavelength 760 nm. Hence, wavelength of the second photon is:

A. 460 nm

B. 1060 nm

C. 496 nm

D. 300 nm

Question

Electrons are emitted with zero velocity from a metal surface when it is exposed to radiation of wavelength 6800 Å. Calculate threshold frequency and work function (W_0) of the metal.

Question

The threshold frequency for a metal is $7.0 \times 10^{14}\ s^{-1}$. Calculate the kinetic energy of an electron emitted when radiation of frequency $= 1.0 \times 10^{15}\ s^{-1}$ hits the metal.

Question

A photon of wavelength 4×10^{-7} m strikes on metal surface, the work function of the metal being 2.13 eV. Calculate

(i) the energy of the photon (eV),

(ii) the kinetic energy of the emission, and

(iii) the velocity of the photoelectron ($1 \text{ eV} = 1.6020 \times 10^{-19}$ J).

Question

The threshold frequency for the ejection of electrons from potassium metal is 5.3×10^{14} s^{-1}. Will the photon of a radiation having energy 3.3×10^{-19} J exhibit photoelectric effect? ($h = 6.626 \times 10^{-34}$ Js).

A. Occur

B. Not occur

C. Cannot decide

D. Both A and B

Question

The threshold frequency n_0 for a metal is $6 \times 10^{14}\ s^{-1}$. Calculate the kinetic energy of an electron emitted when radiation of frequency = $1.1 \times 10^{15}\ s^{-1}$ hits the metal.

A. 3.313×10^{-19} J

B. 2.13×10^{-19} J

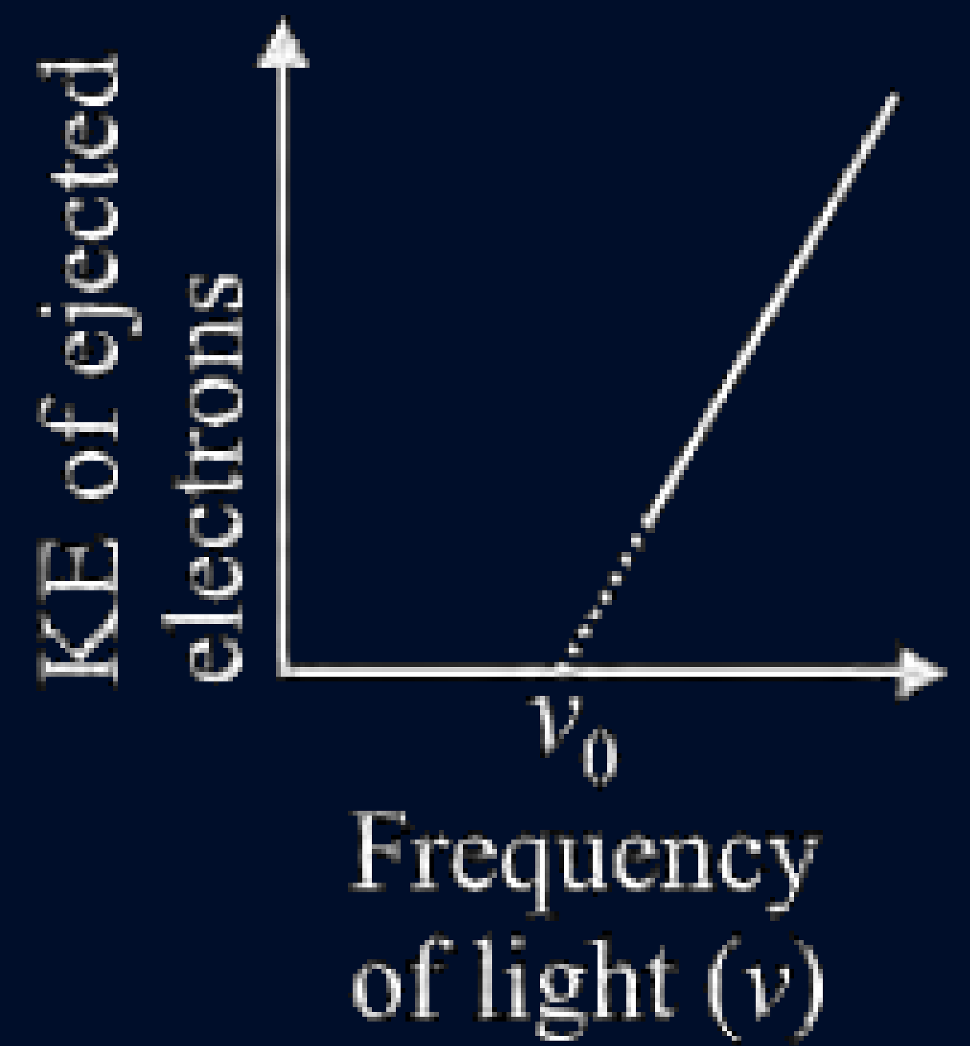
C. 5.53×10^{-19} J

D. 3.313×10^{-14} J

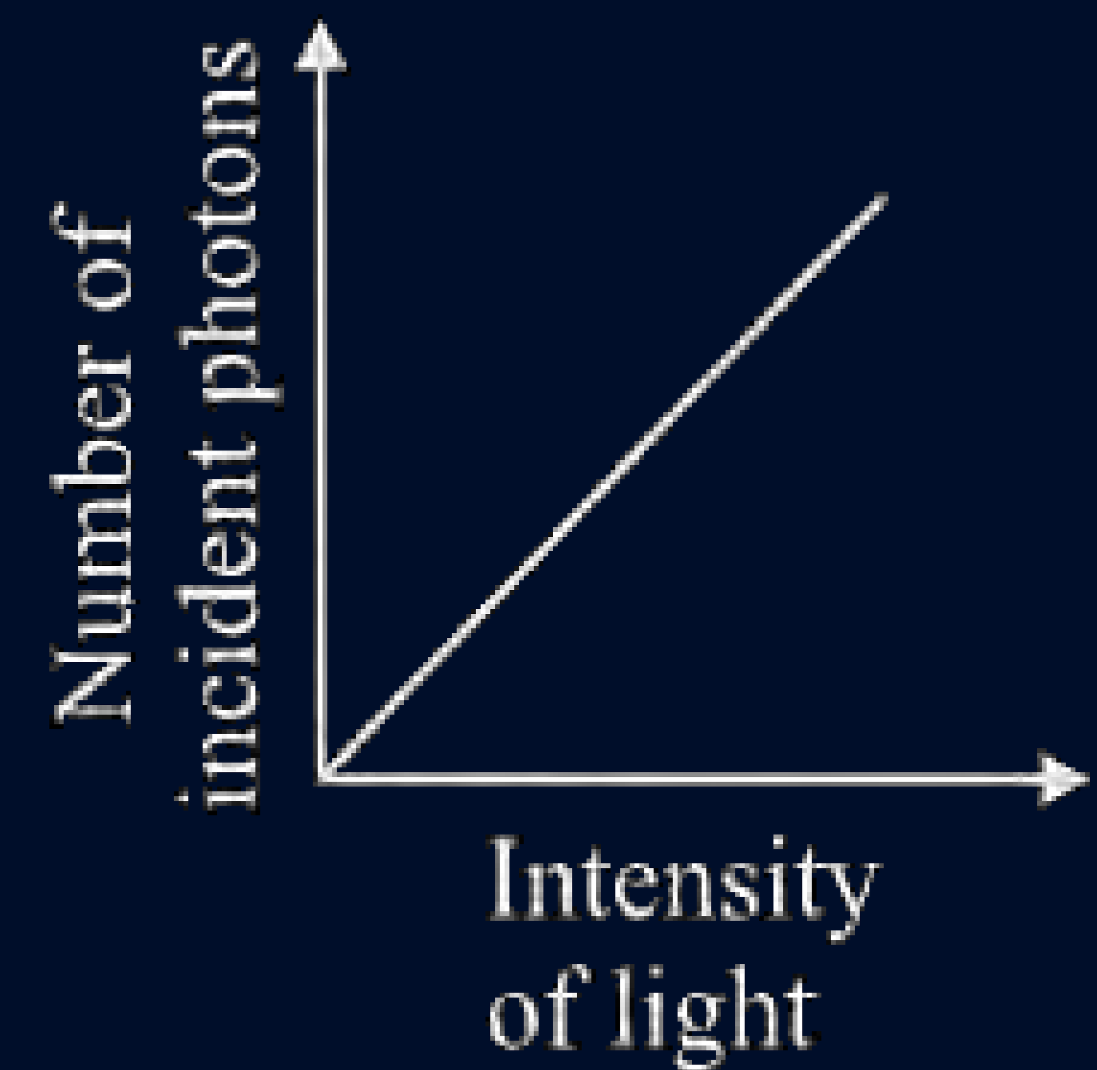
Question

Which of the following is an incorrect graphical representation based on photoelectric effect?

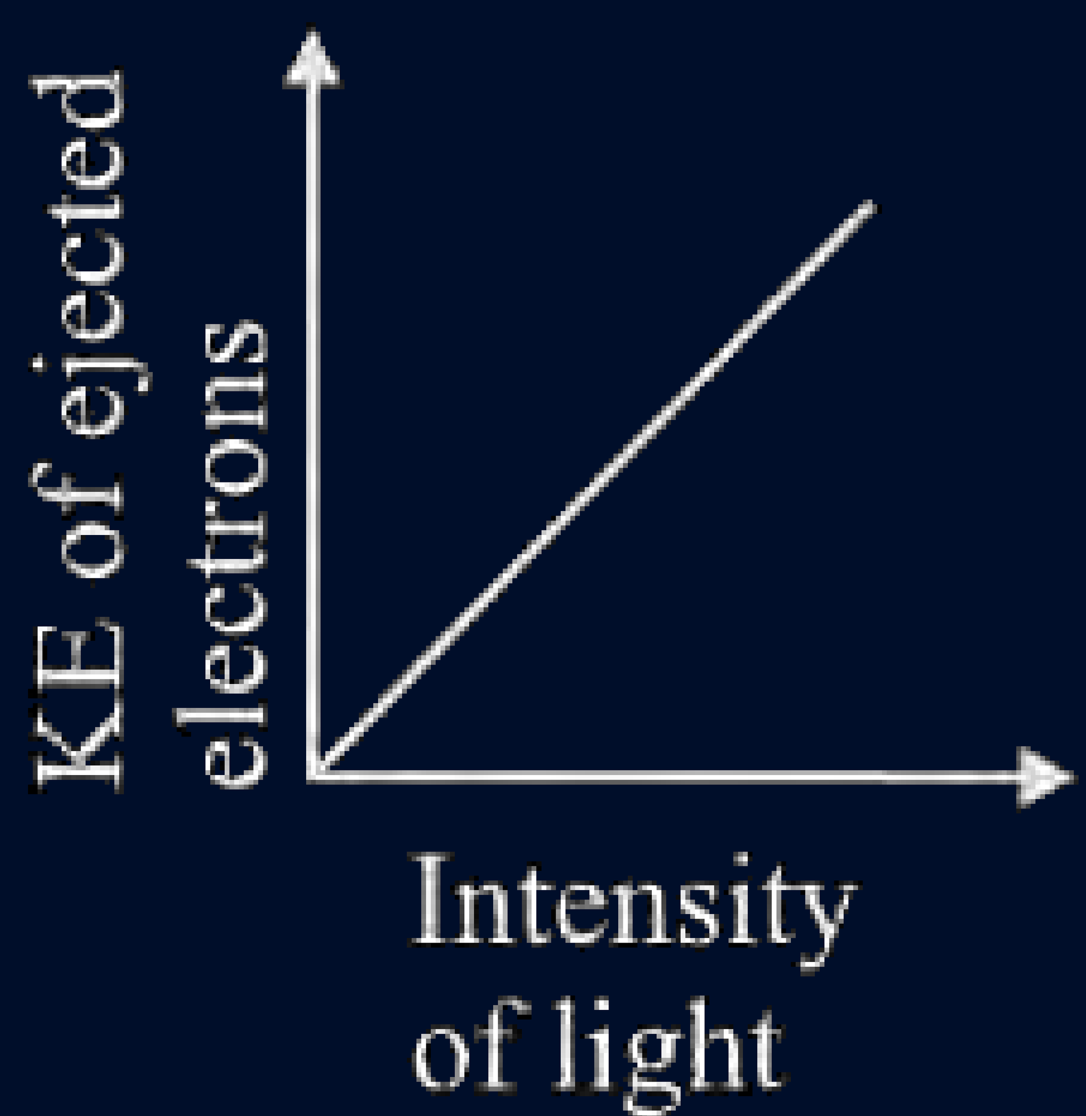
A



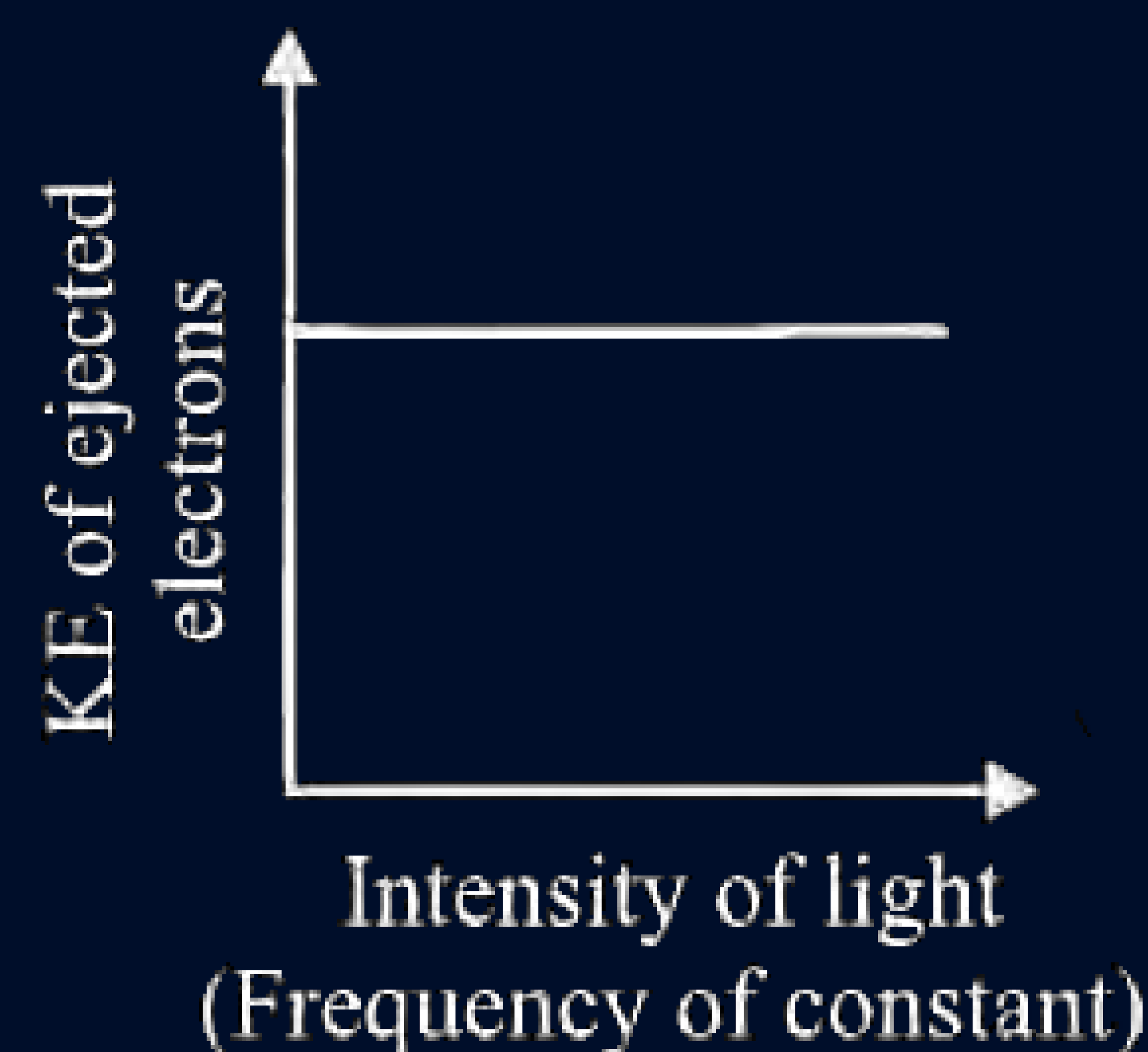
B



C



D



Question

When the frequency of the incident radiation on a metallic plate is doubled, KE of the photoelectrons will be

A. Doubled

B. Halved

C. More than doubled

D. Increases but less than doubled

Question

In photoelectric effect the number of photo-electrons emitted is proportional to:

(A) Intensity of incident beam

(B) Frequency of incident beam

(C) Velocity of incident beam

(D) Work function of photo-cathode

Question

Photoelectric emission is observed from a metal surface with incident frequencies v_1 and v_2 where $v_1 > v_2$. If the kinetic energies of the photoelectrons emitted in the two cases are in the ratio 2 : 1, then the threshold frequency v_0 of the metal is

A. $v_1 > v_2$

B. $\frac{v_1 - v_2}{h}$

C. $2v_1 - v_2$

D. $2v_2 - v_1$

Question

Which of the given statements correctly represents the effect of rise in temperature on the emitted radiations of a hot body?

A. The radiations move towards shorter wavelengths

B. The radiations move towards shorter frequency

C. The radiations move towards lower energy

D. The frequency of radiations does not change

Question

Kinetic energy of the ejected electron is:

A. equal to the frequency of the electromagnetic radiation

B. proportional to the frequency of the electromagnetic radiation

C. more than the frequency of the electromagnetic radiation

D. inversely proportional to the frequency of the electromagnetic radiation

Question

Which of the following is not a permissible value of angular momentum of electron in H-atom?

(A) $1.5\frac{h}{\pi}$

(B) $0.5\frac{h}{\pi}$

(C) $1.25\frac{h}{\pi}$

(D) All of these

Question

Which of the following particle is not deflected in the magnetic field?

A. Electron

B. Proton

C. Neutron

D. Deuteron

Question

Which of the following electromagnetic radiation have greater frequency?

A. X-rays

B. Ultraviolet rays

C. Radio waves

D. Visible rays

Question

The energy of a radiation of wavelength 8000 Å is E_1 and energy of a radiation of wavelength 16000 Å is E_2. What is the relation between these two

A. $E_1 = 6E_2$

B. $E_1 = 2E_2$

C. $E_1 = 4E_2$

D. $E_1 = \frac{1}{2} E_2$

Question

The energies E_1 and E_2 of two radiations are 25 eV, and 50 eV, respectively. The relation between their wavelengths, i.e. λ_1 and λ_2 will be

A $\lambda_1 = \frac{1}{2}\lambda_2$

B $\lambda_1 = \lambda_2$

C $\lambda_1 = 2\lambda_2$

D $\lambda_1 = 4\lambda_2$

Question

Number of photons emitted by a 100 W (Js^{-1}) yellow lamp in 1.0 s is (λ of yellow light is 560 nm)

(A) 1.6×10^{18}

(B) 1.4×10^{18}

(C) 2.8×10^{20}

(D) 2.1×10^{20}

Question

The work function of a metal is 4.2 eV. If radiation of 2000 Å fall on the metal then the kinetic energy of the fastest photoelectron is:

(A) 1.6×10^{-19} J

(B) 16×10^{-10} J

(C) 3.2×10^{-19} J

(D) 6.4×10^{-10} J

Question

Bohr's model may be applied to

A. Na^{10+} ion

B. He atom

C. Be^{2+} ion

D. C^{6+} ion

Question

If r_1 is the radius of the first orbit of hydrogen atom, then the radii of second, third and fourth orbits in terms of r_1 are.

A. r_1^2, r_1^3, r_1^4

B. $8r_1, 27r_1, 64r_1$

C. $4r_1, 9r_1, 16r_1$

D. $2r_1, 6r_1, 8r_1$

Question

The ratio of circumference of third and second orbits of He^+ ion is:

A) 3 : 2

B) 2 : 3

C) 9 : 4

D) 4 : 9

Question

If the radius of first orbit of H-atom is x Å, then the radius of the second orbit of Li^{2+} ion will be:

(A) x Å

(B) $\frac{4x}{3}$ Å

(C) $\frac{9x}{2}$ Å

(D) 4x Å

Question

If velocity of an electron in Ist orbit of H atom is V, what will be the velocity of electron in 3rd orbit of Li^{+2}:

(A) V

(B) V/3

(C) 3 V

(D) 9 V

Question

The ratio of time taken by electron in revolutions round the H-nucleus in the second and third orbits is:

A. 2 : 3

B. 4 : 8

C. 8 : 27

D. 27 : 8

Question

In which of the following radius of the first orbit is minimum?

A. A Hydrogen atom

B. A tritium atom

C. Triply ionized beryllium

D. Double ionized helium

Question

Energy of orbit

A. Increases as we move away from nucleus

B. Decreases as we move away from nucleus

C. Remains same as we move away from nucleus

D. None of these

Question

Magnitude of K.E. in an orbit is equal to

A. Half of the potential energy

B. Twice of the potential energy

C. One fourth of the potential energy

D. None of these

Question

If the total energy of an electron is -1.51 eV in hydrogen atom then find out K.E, P.E, orbit's radius and velocity of the electron in that orbit.

Question

The value of the energy for the first excited state of hydrogen atom will be

(A) -13.6 eV

(B) -3.40 eV

(C) -1.51 eV

(D) -0.85 eV

Question

The ratio of energies of first excited state of He^+ ion and ground state of H–atom is:

(A) 1 : 1

(B) 4 : 1

(C) 1 : 4

(D) 16 : 1

Question

Calculate the radius of 3rd excited state of Li^{+2}.

A) 2.82 Å

B) 4.52 Å

C) 7.23 Å

D) 1.75 Å

Question

Calculate the radius ratio of 2^{nd} excited state of H and 1^{st} excited state of Li^{+2}.

A) 9 : 4

B) 27 : 4

C) 16 : 9

D) 25 : 9

Question

If the total energy of an electron is 0.54 eV in hydrogen atom then find out P.E. of the electron in that orbit

(A) –1.08 eV

(B) 0.27 eV

(C) –0.54 eV

(D) 2.16 eV

Question

The energy needed to excite a hydrogen atom from its ground to its third excited state is:

A. 12.1 eV

B. 10.2 eV

C. 0.85 eV

D. 12.75 eV

Question

The ionisation energy of a hydrogen atom is 13.6 ev. The energy of the ground level in doubly ionised lithium is:

A. 28.7 eV

B. 54.4 eV

C. –122.4 eV

D. 13.6 eV

Question

How much energy would be required by an electron while moving from ground state to 3rd excited state of He^{+} ion?

(A) 40.8 eV

(B) 10.2 eV

(C) 51 eV

(D) 48.35 eV

Question

Calculate the ratio of energies of He^+ for 1^{st} and 2^{nd} excited state.

Question

The ionization energy for the hydrogen atom is 13.6 eV then the required energy in eV to excite it from the ground state to 1st excited state

Question

Wavelength of the first line of Paschen series is

A [18750] Å

B [2854] Å

C [3452] Å

D [6243] Å

Question

The number of possible spectral lines in Brackett series in hydrogen spectrum, when the electrons present in the ninth excited state return to the ground state, is:

A 36

B 45

C 5

D 6

Question

In hydrogen spectrum, the series of lines appearing in ultraviolet region of electromagnetic spectrum are called

(A) Lyman lines

(B) Balmer lines

(C) Pund lines

(D) Brackett lines

Question

Which of the following series of lines in the atomic spectrum of hydrogen appear in the visible region?

A. Lyman

B. Paschen

C. Brackett

D. Balmer

Question

When electron jumps from the fourth orbit to the second orbit in He^+ ion, the radiation emitted out will fall in

A. Ultraviolet region

B. Visible region

C. Infrared region

D. Radio wave region

Question

To which electronic transition between Bohr orbits in hydrogen, the second line in the Balmer series belongs?

A. 3 to 2

B. 4 to 2

C. 5 to 2

D. 6 to 2

Question

Calculate wavelength of 3rd line of Brackett series in hydrogen spectrum

A) $\frac{784}{33R}$

B) $\frac{33R}{784}$

C) $\frac{784R}{33}$

D) $\frac{33}{784R}$

Question

What is the maximum wavelength line in the Lyman series of He ion?

A) $3R$

B) $\frac{1}{3R}$

C) $\frac{4}{4R}$

D) None of these

Question

Ratio between longest wavelength of H atom in Lyman series to the shortest wavelength in Balmer series of He^+ is:

A. 4/3

B. 36/5

C. 1/4

D. 5/9

Question

$^{14}_{6}C$ and $^{14}_{7}N$ are the examples of

A. isobars

B. Isotopes

C. isotones

D. None of these

Question

99.985% of hydrogen atoms contain only one proton. This isotope is called

A. protium

B. deuterium

C. tritium

D. Both (A) and (B)

Question

Calculate the number of protons, neutrons and electrons in $^{80}_{35}Br$.

A. Protons = 80, electrons = 80, neutrons = 35

B. Protons = 35, electrons = 55, neutrons = 80

C. Protons = 35, electrons = 35, neutrons = 80

D. Protons = 35, electrons = 35, neutrons = 45

Question

The value 109677 cm^{-1} is called

A. Pfund constant

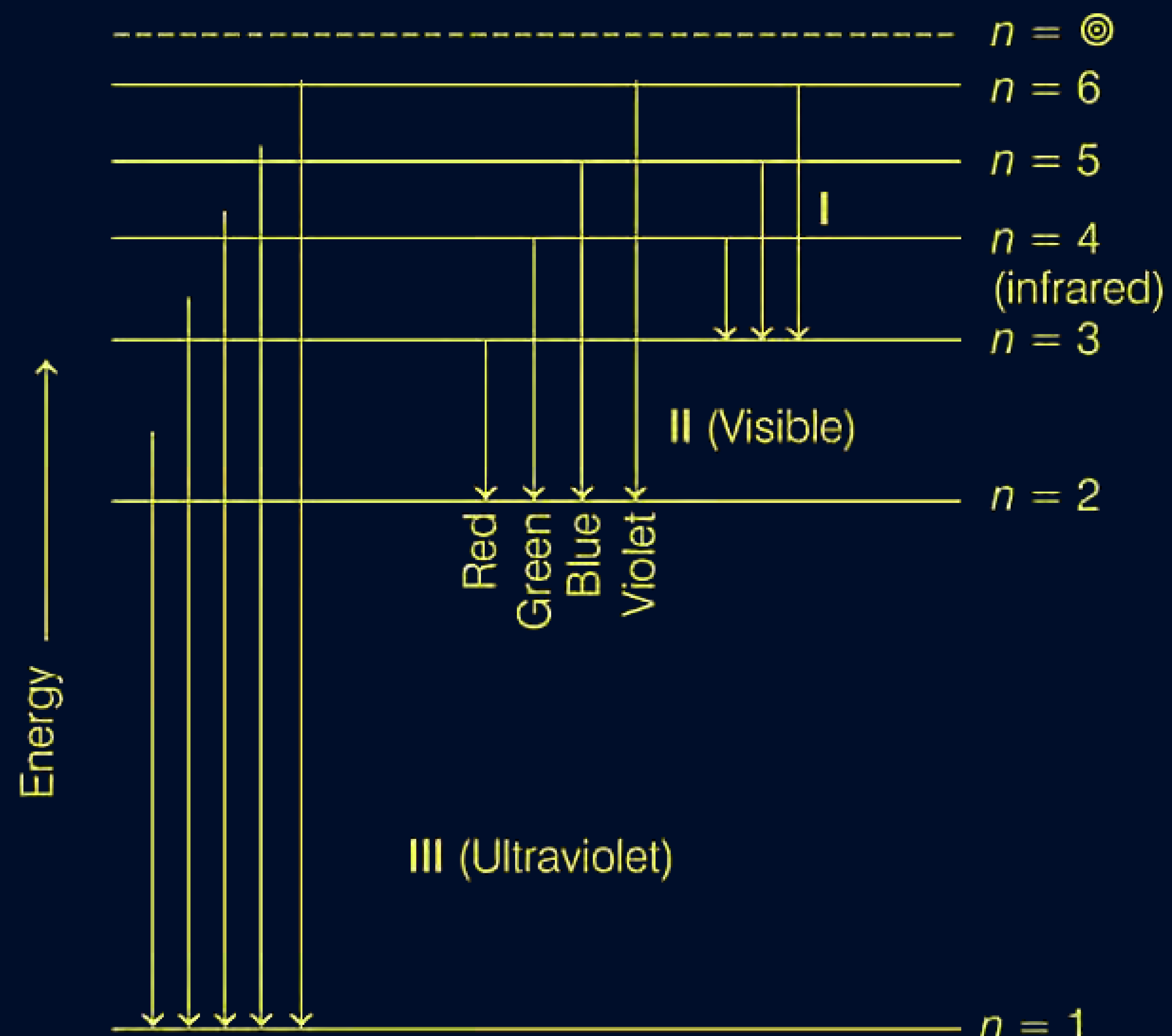
B. Balmer constant

C. Rydberg constant

D. Lyman constant

Question

Transition of the electron in the hydrogen atom is shown in the given figure.



Identify I, II, III in the above figure.

A. I → Lyman series; II → Paschen series; III → series Balmer

B. I → Paschen series; II → series Lyman series; III → Balmer

C. I → Paschen series; II → Balmer series; III → Lyman series

D. I → Balmer series; II → Lyman series; III → Paschen series

Question

The electronic transition from $n = 2$ to $n = 1$ will produce the shortest wavelength in (where, n = principal quantum number)

A. He^+

B. H

C. H^+

D. Li^{2+}

Question

Which of the following relation correctly described all the lines in the hydrogen spectrum?

A. $\bar{v} = \frac{1}{109677}\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]cm^{-1}$

B. $\bar{v} = 109677\left[\frac{1}{n_2^2} - \frac{1}{n_1^2}\right]cm^{-1}$

C. $\bar{v} = 109677[n_1^2 - n_2^2]cm^{-1}$

D. $\bar{v} = 109677[n_2^2 - n_1^2]cm^{-1}$

Question

Zeeman effect is due to

A splitting up the lines in an emission spectrum in the presence of an external electrostatic field

B random scattering of light by colloidal particles

C splitting up of the lines in an emission spectrum in a magnetic field

D emission of electrons from metals when light falls upon them

Question

Splitting of spectral lines when atoms are subjected to strong electric field is called:

A. Zeeman effect

B. Stark effect

C. Decay

D. Disintegration

Question

Minimum de-Broglie wavelength is associated with

(A) Electron

(B) Proton

(C) CO_2 molecule

(D) SO_2 molecule

Question

Which of the following matter waves will have the shortest wavelength, if travelling with same kinetic energy?

A. Electron

B. α-particle

C. Neutron

D. Proton

Question

The de Broglie wavelength of a tennis ball mass 60 g moving with a velocity of 10 m per second is approximately:

A. 10^{-16} m

B. 10^{-25} m

C. 10^{-33} m

D. 10^{-31} m

Question

If wavelength is equal to the distance traveled by the electron in one second, then:

A $\lambda = \frac{h}{p}$

B $\lambda = \frac{h}{m}$

C $\lambda = \sqrt{\frac{h}{p}}$

D $\lambda = \sqrt{\frac{h}{m}}$

Question

An electron is moving with a kinetic energy of 4.55×10^{-25} J. What will be de Broglie wavelength for this electron?

A) 5.28×10^{-7} m

B) 7.28×10^{-7} m

C) 2×10^{-10} m

D) 3×10^{-5} m

Question

In H-atom, if 'x' is the radius of the first Bohr orbit, de Broglie wavelength of an electronic 3rd orbit is

(A) $3\pi x$

(B) $6\pi x$

(C) $\frac{9x}{2}$

(D) $\frac{x}{2}$

Question

What possibly can be the ratio of the de-Broglie wavelengths for two electrons each having zero initial energy and accelerated through 50 volts and 200 volts

A. 3 : 10

B. 10 : 3

C. 1 : 2

D. 2 : 1

Question

Simultaneous determination of exact position and momentum of an electron is:

A. Possible

B. Impossible

C. Sometimes possible sometimes impossible

D. None of the above

Question

Uncertainty in position is twice the uncertainty in momentum. Uncertainty in velocity is:

A $\sqrt{\frac{\hbar}{\pi}}$

B $\frac{1}{2m}\sqrt{\frac{\hbar}{\pi}}$

C $\frac{1}{2m}\sqrt{\hbar}$

D $\frac{\hbar}{4\pi}$

Question

Uncertainty in position of a 0.25 g particle is 10^{-5}. Uncertainty of velocity is ($h = 6.6 \times 10^{-34}$ J s)

A) 1.2×10^{34}

B) 2.1×10^{-26}

C) 1.6×10^{-20}

D) 1.7×10^{-9}

Question

A microscope using suitable photons is employed to locate an electron in an atom within a distance of 0.1 Å. What is the uncertainty involved in the measurement of its velocity?

(A) $2.69 \times 10^{6}\ ms^{-1}$

(B) $5.79 \times 10^{5}\ ms^{-1}$

(C) $5.79 \times 10^{6}\ ms^{-1}$

(D) $4.62 \times 10^{6}\ ms^{-1}$

Question

A golf ball has a mass of 40 g, and a speed of 45ms^{-1}. If the speed can be measured with in accuracy of 2%, calculate the uncertainty in the position.

A. 2.52×10^{-33} m

B. 3.02×10^{-33} m

C. 1.46×10^{-33} m

D. 0.82×10^{-33} m

Question

If the kinetic energy of an electron is increased 4 times, the wavelength of the de-Broglie wave associated with it would become:

A. four times

B. two times

C. half times

D. one fourth times

Question

No. of waves in fourth orbit is:

A 4

B 5

C 0

D 1

Question

If uncertainty in position of an e^- is same as the Δx of He atom. If Δp of e^- is 32×10^5 then find Δp in He atom:

(A) 32×10^5

(B) 18×10^5

(C) 16×10^5

(D) 64×10^5

Question

If the uncertainty in velocity and position is same, then the uncertainty in momentum would be:

(A) $\sqrt{\frac{hm}{4\pi}}$

(B) $m\sqrt{\frac{h}{4\pi}}$

(C) $\sqrt{\frac{h}{4\pi m}}$

(D) $\frac{1}{m}\sqrt{\frac{h}{4\pi}}$

Question

Heisenberg's Uncertainty principle is not valid for

(A) Moving electron

(B) Motor car

(C) Stationary particles

(D) (B) and (C) both

Question

If the uncertainty in position of a moving particle is 0 then find out ΔP

A) 0

B) 1

C) Infinite

D) None of these

Question

Uncertainty in the position of an electron (mass = 9.1×10^{-31} kg) moving with a velocity 300 m/s accurate up to 0.001% will be

A. 5.76×10^{-3} m

B. 1.92×10^{-2} m

C. 3.84×10^{-3} m

D. 19.2×10^{-4} m

Question

Uncertainty in measuring the speed of a particle is numerically equal to the uncertainty in measuring its position. The value of these uncertainties will be

(A) Equal to $\sqrt{\frac{h}{4\pi m}}$

(B) Less than $\sqrt{\frac{h}{4\pi m}}$

(C) Greater than $\sqrt{\frac{h}{4\pi m}}$

(D) Both (A) or (C)

Question

The effect of Heisenberg uncertainty principle is significant

A. only for motion of microscopic objects

B. negligible for that of macroscopic objects

C. Both (A) and (B)

D. None of the above

Question

Which of the following is the correct form of Schrodinger wave equation?

(A) $$\frac{\partial^2\psi}{\partial^2x}+\frac{\partial^2\psi}{\partial^2y}+\frac{\partial^2\psi}{\partial^2z}+\frac{16\pi^2m}{h^2}(E-V)\psi=0$$

(B) $$\frac{\partial^2\psi}{\partial^2x}+\frac{\partial^2\psi}{\partial^2y}+\frac{\partial^2\psi}{\partial^2z}+\frac{8\pi^2m}{h^2}(E-V)\psi=0$$

(C) $$\frac{\partial^2\psi}{\partial^2x}+\frac{\partial^2\psi}{\partial^2y}+\frac{\partial^2\psi}{\partial^2z}+\frac{4\pi^2m}{h^2}(E-V)\psi=0$$

(D) $$\frac{\partial^2\psi}{\partial^2x}+\frac{\partial^2\psi}{\partial^2y}+\frac{\partial^2\psi}{\partial^2z}+\frac{8\pi^2m^2}{h^2}(E-V)\psi=0$$

Question

Which of the following orbital is the complete wave function represented by Ψ_{410}?

(A) 4s

(B) 3p

(C) 4p

(D) 4d

Question

Which orbital is represented by $\Psi_{4,2,0}$

A. 4d

B. 3d

C. 4p

D. 4s

Question

The orbital angular momentum of an electron is 2s orbital is

(A) $+\frac{1}{2} \cdot \frac{h}{2\pi}$

(B) 0

(C) $\frac{h}{2\pi}$

(D) $\sqrt{2}\frac{h}{2\pi}$

Question

The orbital angular momentum of a 4p electron will be

A) $4 \cdot \frac{h}{2\pi}$

B) $\sqrt{2} \cdot \frac{h}{2\pi}$

C) $\sqrt{6} \cdot \frac{h}{2\pi}$

D) $\sqrt{2} \cdot \frac{h}{4\pi}$

Question

A subshell $n = 5$, $l = 3$ can accommodate:

A. 10 electrons

B. 14 electrons

C. 18 electrons

D. None of these

Question

If $\ell = 3$ then type of subshell and number of orbital is

A. 3p, 3

B. 4f, 14

C. 5f, 7

D. 3d, 5

Question

Spin angular momentum for electron

(A) $\sqrt{s(s+1)}\dfrac{h}{2\pi}$

(B) $\sqrt{2s(s+1)}\dfrac{h}{2\pi}$

(C) $\sqrt{s(s+2)}\dfrac{h}{2\pi}$

(D) None

Question

What will be the orbital angular momentum of 3d orbital

(A) $\sqrt{8}.\dfrac{h}{2\pi}$

(B) $\sqrt{12}.\dfrac{h}{2\pi}$

(C) $\sqrt{3}\dfrac{h}{2\pi}$

(D) $\sqrt{6}.\dfrac{h}{2\pi}$

Question

How many total nodes are there in 4p:

A) 2

B) 3

C) 4

D) 0

Question

Which of the following set of quantum numbers is not possible?

A. a

B. b

C. c and d

D. c

(a) $n = 2, l = 1, m = 0, s = -\frac{1}{2}$

(b) $n = 3, \ell = 2, m = 0, s = \pm\frac{1}{2}$

(c) $n = 2, l = 3, m = -2, s = \pm\frac{1}{2}$

(d) $n = 4, \ell = 2, m = -3, s = \pm\frac{1}{2}$

Question

Two electrons occupying the same orbital are distinguished by

A. Magnetic quantum number

B. Azimuthal quantum number

C. Spin quantum number

D. Principal quantum number

Question

In potassium the order of energy level for 19^{th} electron is:

A) $3s > 3d$

B) $4s < 3d$

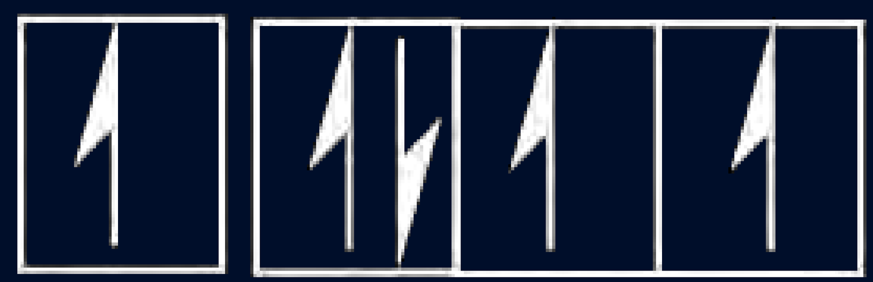
C) $4s > 4p$

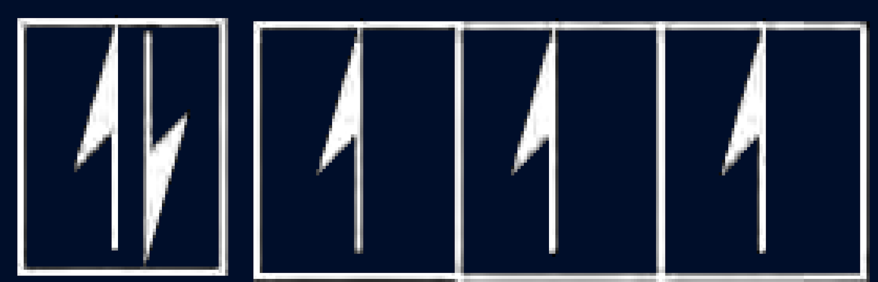
D) $4s = 3d$

Question

The orbital diagram in which aufbau principle is violated is:

(A) 

(B) 

(C) 

(D) 

Question

According to which rule electronic configuration of Cr is $[Ar]^3d5^4s^1$ in place of $[Ar]^3d4^4s^2$

A. Pauli's Exclusion Principle

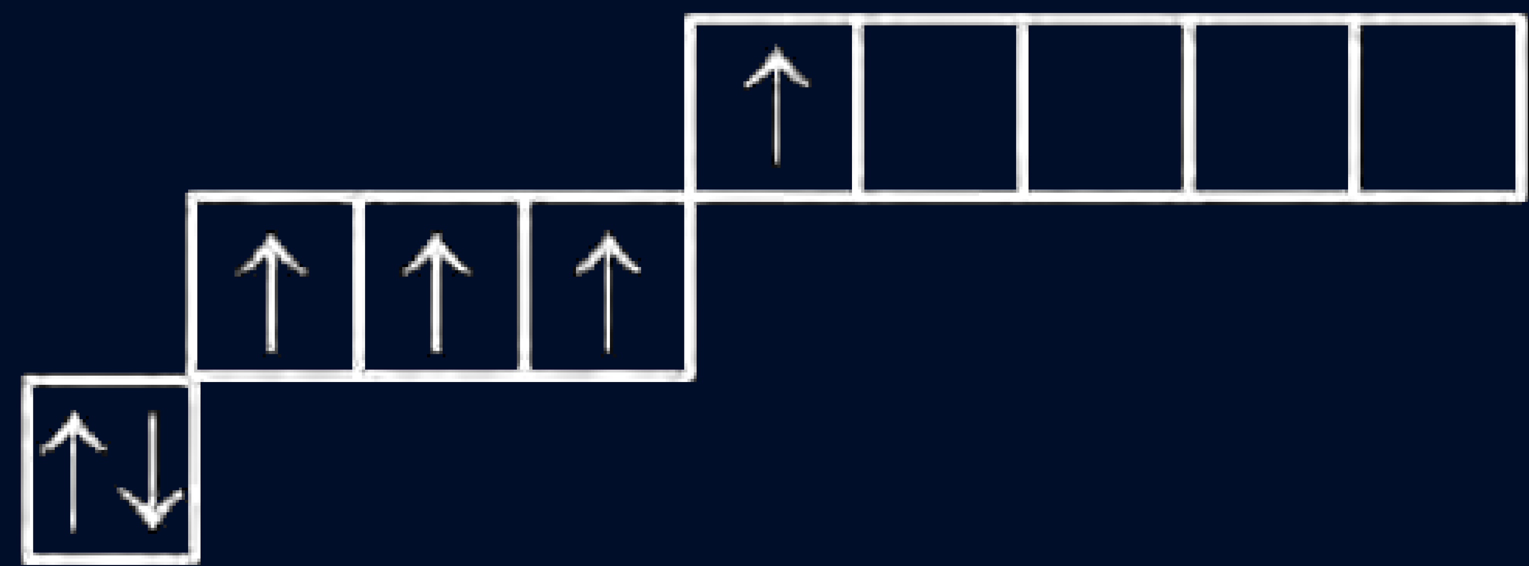
B. Hund's Rule

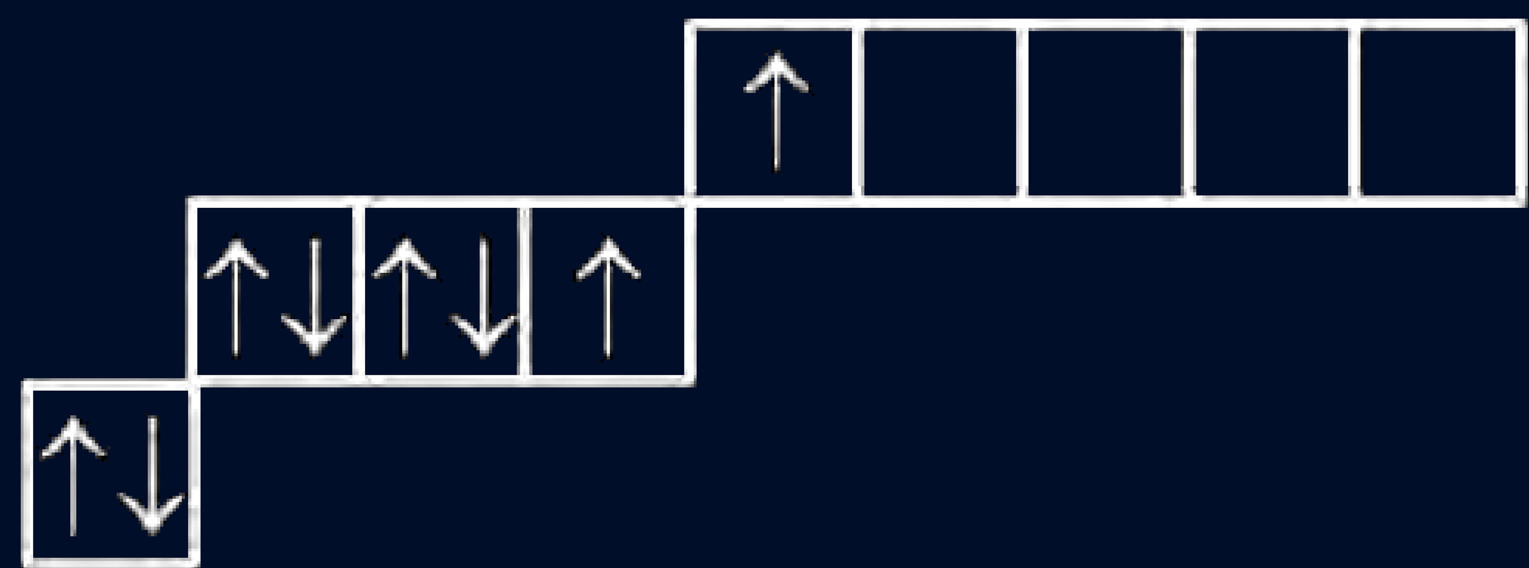
C. Afbau Principle

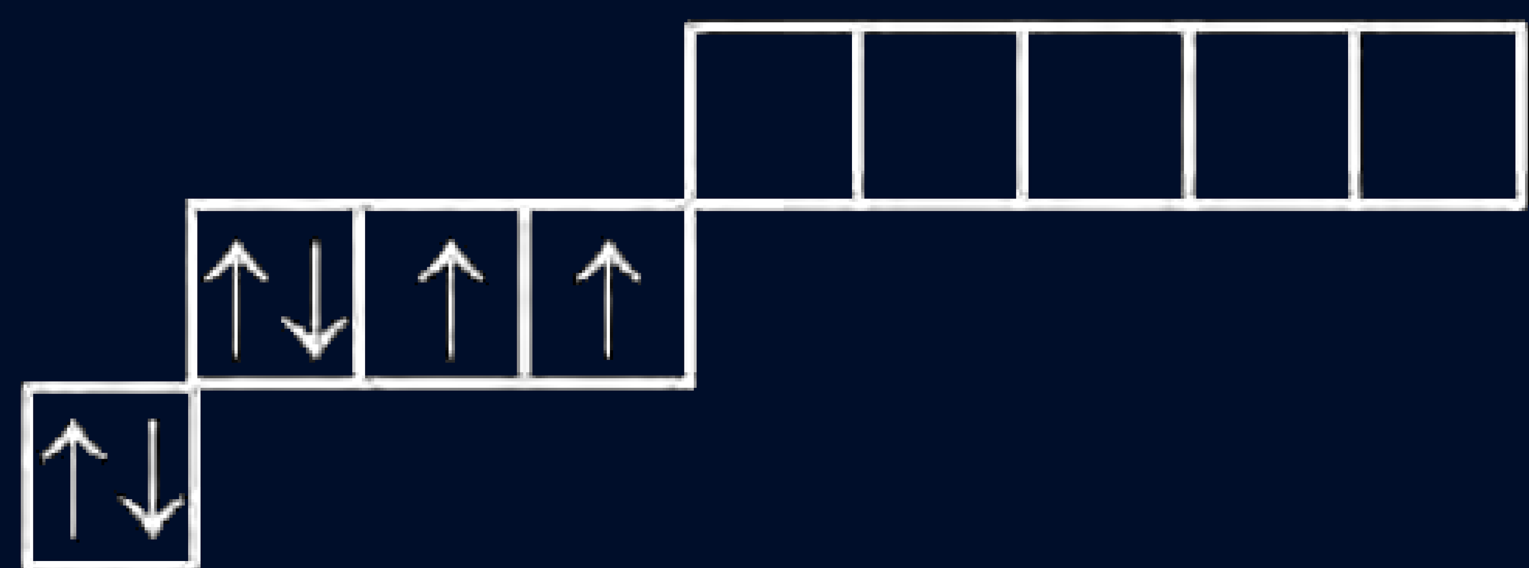
D. Heisenberg Uncertainty Principle

Question

Which one represent ground state configuration?

A 

B 

C 

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